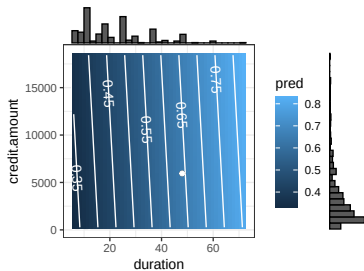
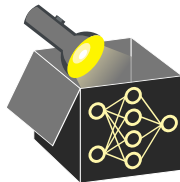


Interpretable Machine Learning

Local Explanations: Lime Examples



Learning goals

- See real-world data examples
- See application to image and text data

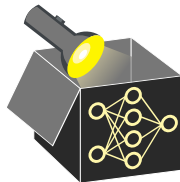
EXAMPLE: CREDIT SCORING (TABULAR DATA)

- **Black-box model** $\hat{f}_{bad}$: SVM with RBF kernel (predicts probability of bad credit risk)

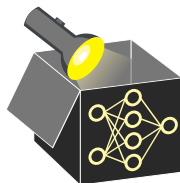
- **Instance to explain $\mathbf{x}$** : First row in the dataset, with $\hat{f}_{bad}(\mathbf{x}) = 0.658$

duration	sex	credit.amount	purpose	housing	age	saving	checking	...
48	female	5951	radio/TV	own	22	little	moderate	...

- **Surrogate model**: LASSO, restricted to 5 non-0 feats (via regularization)
- **Training data for surrogate**: Samples $\mathbf{z}$, weighted by Gower dist. to $\mathbf{x}$



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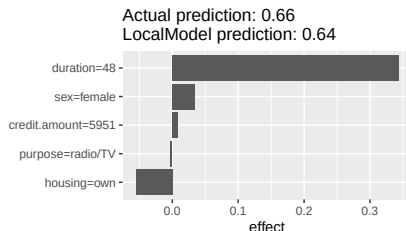
- **Surrogate model**: LASSO, restricted to 5 non-0 feats (via regularization)
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- **Prediction:**

$$\hat{g}(\mathbf{x}) = 0.640 \text{ vs. } \hat{f}_{bad}(\mathbf{x}) = 0.658$$

↪ $\hat{g}$ provides good local approx. of $\hat{f}_{bad}$, but omits several features

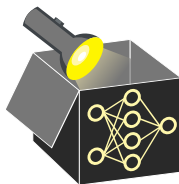
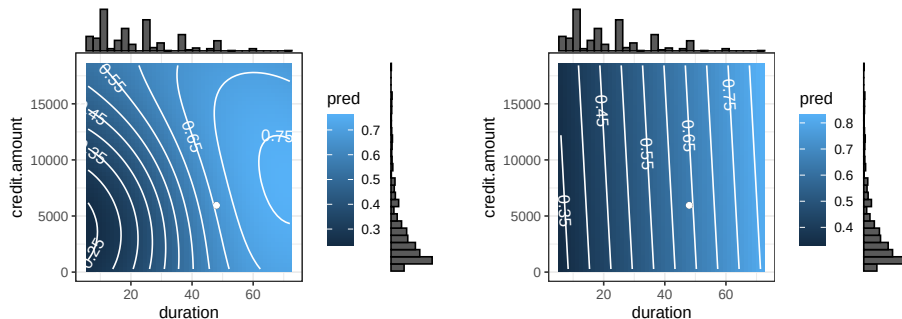
↪ Small mismatch reflects trade-off:
interpretability vs. fidelity



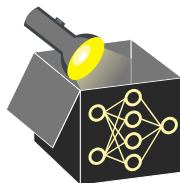
Interpretation: Prediction is mainly driven by loan duration, with small positive effect from sex and credit.amount, and negative contributions from housing and purpose.

EXAMPLE ON CREDIT DATASET (CONT'D)

- 2D ICE plots (pred. surface plots) for duration and credit.amount
- Illustration how $\hat{g}$ linearly approximates nonlinear decision surface of $\hat{f}_{bad}$



- **Left:** 2D ICE plot of $\hat{f}_{bad}$ showing decision surface
- **Right:** Linear approximation by surrogate model $\hat{g}$.
 - ↪ White dot indicates input $\mathbf{x}$ to be explained
 - ↪ Histograms show marginal distribution of features in training data



LIME can also be applied to text data:

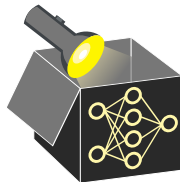
- Raw text representations:
 - Binary vector indicating the presence or absence of a word
 - A vector of word counts
- Examples for *“This text is the first text.”* and *“Finally, this is the last one.”*:

this	text	is	the	first	finally	last	one
1	2	1	1	1	0	0	0
1	0	1	1	0	1	1	1

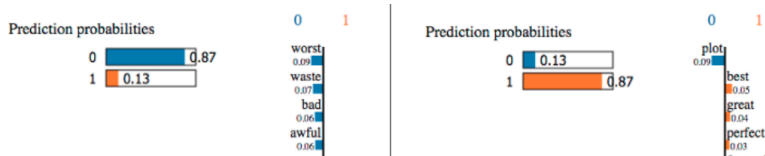
- **Sampling:** Randomly set the entry of individual words to 0; equal to removing all occurrences of this word in the text.
- **Proximity:** Exponential kernel with cosine distance.
 - Neglects words that do not occur in both texts
 - Measures the distance irrespective of the text size

LIME FOR TEXT DATA (CONT'D)

► “Shen, Ian,” 2019



- Random forest classifier labeling movie reviews from IMDB
 - 0: negative
 - 1: positive
- Surrogate model is a sparse linear model



Words like “worst” or “waste” indicate negative review while words like “best” or “great” indicate positive review

LIME FOR IMAGE DATA

LIME also works for image data:

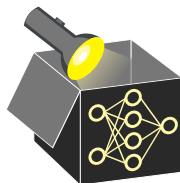
- **Idea:** Each obs. is represented by a binary vector indicating the presence or absence of superpixels

► "Achanta et al." 2012

- Superpixels are interconnected pixels with similar colors (absence of a single pixel might not have a (strong) effect on the prediction)
- **Warning:** Size of superpixels needs to be determined before the segmentation takes place
- **Sampling:** Randomly switching some of the superpixels "off", i.e., by coloring some superpixels uniformly



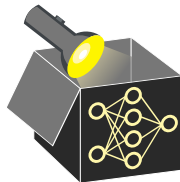
Example for
superpixels of different
sizes



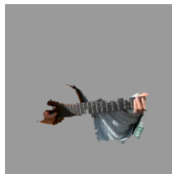
LIME FOR IMAGE DATA (CONT'D)

► "Ribeiro." 2016

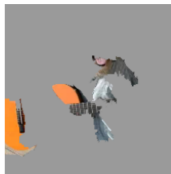
- Explaining prediction of pre-trained inception neural network classifier
- **Sampling**: Graying out all superpixels besides 10 superpixels
- **Surrogate**: Locally weighted sparse linear models
- **Proximity**: Exponential kernel with euclidean distance



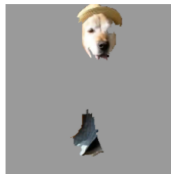
(a) Original Image



(b) Explaining *Electric guitar*



(c) Explaining *Acoustic guitar*



(d) Explaining *Labrador*

Top 3 classes predicted